M3403 WEEK 2 2020/21

6. BERNOULLI EQUATION

These are equations with the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \text{, where } n \text{ is a real number}$$

N.B If $n=1$ or $n=0$, we have a linear equation

To solve Bernoulli equation, we make the substitution $u = y^{1-n}$, which transforms the equation into a linear equation.

Examples

Solve ① $\frac{dy}{dx} + \frac{y}{x} = x^2y^2$

② $x\frac{dy}{dx} + y = \frac{1}{y^2}$

③ $\frac{dy}{dx} + \frac{1}{3}y = e^x y^4$

Soln ① $\frac{dy}{dx} + \frac{y}{x} = x^2y^2$

$\Rightarrow y^{-2}\frac{dy}{dx} + y^{-1}x^{-1} = x^2$

Let $v = y^{-1}$ $\Rightarrow$ $-\frac{dv}{dx} = y^{-2}\frac{dy}{dx}$

$\frac{dv}{dx} - vx^{-1} = -x^2$ which is linear

Integrating factor $= \exp\left(\int -\frac{1}{x}\,dx\right)$

$= \frac{1}{x}$

$$\frac{1}{x}\frac{dv}{dx} - \frac{v}{x^2} = -\frac{x^2}{x}$$

$$\Rightarrow \frac{d}{dx}(vx^{-1}) = -x$$

$$\Rightarrow vx^{-1} = -\frac{x^2}{2} + C$$

$$\Rightarrow \frac{1}{xy} = -\frac{x^2}{2} + C$$

$$\Rightarrow y = \frac{2}{Cx - x^3} \quad \text{or} \quad y \equiv 0$$

Since we divided with y, $y=0$ got lost as a soln

(2) $$x\frac{dy}{dx} + y = \frac{1}{y^2}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{1}{xy^2}$$

$$\Rightarrow y^2\frac{dy}{dx} + \frac{y^3}{x} = \frac{1}{x}$$

Let $v = y^3 \Rightarrow \frac{1}{3}\frac{dv}{dx} = y^2\frac{dy}{dx}$

$$\frac{1}{3}\frac{dv}{dx} + \frac{v}{x} = \frac{1}{x}$$

$$\Rightarrow \frac{dv}{dx} + \frac{3v}{x} = \frac{3}{x}$$ which is linear

Integrating factor $= \exp\left(\int \frac{3}{x}\,dx\right)$

$= x^3$

$$x^3\frac{dv}{dx} + 3x^2 v = 3x^2$$

$$\Rightarrow \frac{d}{dx}(vx^3) = 3x^2$$

$$\Rightarrow vx^3 = x^3 + C$$

$$\Rightarrow v = \frac{x^3 + C}{x^3}$$

$$\Rightarrow y = \left[1 + \frac{C}{x^3}\right]^{\frac{1}{3}}$$

N.B $y=0$ is not a sol^n since we don't divided by y.

(3) $\frac{dy}{dx} + \frac{1}{3}y = e^x y^4$

$$\Rightarrow y^{-4}\frac{dy}{dx} + \frac{y^{-3}}{3} = e^x$$

let $v = y^{-3} \Rightarrow \frac{dv}{dx} = -3y^{-4}\frac{dy}{dx}$

$$-\frac{dv}{dx} + v = 3e^x$$

$$\Rightarrow \frac{dv}{dx} - v = -3e^x$$ which is linear

Integrating factor $= \exp\left(\int -1\,dx\right)$

$= e^{-x}$

$$e^{-x}\frac{dv}{dx} - ve^{-x} = -3$$

$$\Rightarrow \frac{d}{dx}(ve^{-x}) = -3$$

$$\Rightarrow ve^{-x} = -3x + C$$

$$\Rightarrow y^{-3} e^{-2x} = -3x + C$$

$$\Rightarrow \frac{1}{y^3} = (-3x + C)e^{x}$$

$$\Rightarrow y = \left[\frac{1}{(-3x+C)e^{x}}\right]^{1/3}$$

7. Equations With Linear Coefficients

These are equations of the form

$$(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0 \quad \text{—①}$$

- When $\frac{a_1}{b_1} = \frac{a_2}{b_2}$, eqⁿ ① becomes

$$\frac{dy}{dx} = -\frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$$

$$= \frac{-b_1\left(\frac{a_1}{b_1}x + y\right) + c_1}{b_2\left(\frac{a_1}{b_1}x + y\right) + c_2}$$

which is of the form $\frac{dy}{dx} = F(ax + by)$

- When $\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$ and $c_1 = c_2 = 0$ eqⁿ ① becomes

$$\frac{dy}{dx} = -\frac{a_1x + b_1y}{a_2x + b_2y}$$

which is homogeneous.

- Generally, when $\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$, we make the substitutions $x = u + k$ and $y = v + c$, where k and c are constants satisfying the equations

$$a_1k + b_1c + c_1 = 0$$

$$a_2k + b_2c + c_2 = 0$$

These substitutions transform eqⁿ ① into

$$\frac{dv}{du} = -\frac{a_1u + b_1v}{a_2u + b_2v}$$ which is homogeneous

Examples

Solve ① $(-3x+y-1)dx + (x+y+3)dy = 0$

② $(x+2y-4)dx - (2x-4y)dy = 0$

③ $(x-2y)dx + (2x+y+5)dy = 0$

Soln ① $\frac{dy}{dx} = \frac{3x-y+1}{x+y+3}$

Let $3k - C + 1 = 0 \Rightarrow 3k+1 = C$

$k + C + 3 = 0$

$\Rightarrow k + 3k + 1 + 3 = 0$

$\Rightarrow k = -1 \qquad \Rightarrow C = -2$

Let $x = u - 1$, $y = v - 2$

$\Rightarrow dx = du \qquad \Rightarrow dy = dv$

$$\frac{dv}{du} = \frac{3u - 3 - v + 2 + 1}{u - 1 + v - 2 + 3}$$

$$= \frac{3u - v}{u + v}$$ which is homogeneous

Let $v = tu \Rightarrow \frac{dv}{du} = t + u\frac{dt}{du}$

$$t + u\frac{dt}{du} = \frac{3u - tu}{u + tu} = \frac{3 - t}{t + 1}$$

$$\vdots$$

$$\frac{1+t}{t^2+2t-3}\,dt = -\frac{du}{u}$$

$$\Rightarrow \frac{1+t}{(t+3)(t-1)}\,dt = -\frac{du}{u}$$

$$\Rightarrow \left[\frac{1}{2(t+3)} + \frac{1}{2(t-1)}\right]dt = -\frac{du}{u}$$

$$\Rightarrow \frac{1}{2}\ln|u|t+3| + \frac{1}{2}\ln|t-1| = -\ln u + \ln C$$

$$\Rightarrow \ln\left|u(t+3)(t-1)u^2\right| = \ln C$$

$$\Rightarrow (t+3)(t-1)u^2 = C$$

$$\Rightarrow \frac{y+3x}{u} \cdot \frac{y-x}{u} \cdot u^2 = C$$

$$\Rightarrow (y+2+3x+3)(y+2-x-1) = C$$

$$\Rightarrow (y+3x+5)(y-x+1) = C$$

(2). $(x+2y-4)dx - (2x-4y)dy = 0$

$\Rightarrow (x+2y-4)dx + (4y-2x)dy = 0$

Let $h+2k-4=0$ — ①

$-2h+4k=0 \Rightarrow -h+2k=0$ — ②

Adding ① and ②

$4k-4=0 \Rightarrow k=1$

$\Rightarrow h=2$

let $x = u+2$, $y = v+1$

$dx = du$ $\quad dy = dv$

$$(u+2+2v+2-4)du+(4v+4-2u-4)dv=0$$

$$\Rightarrow \frac{dv}{du} = -\frac{(u+2v)}{4v-2u} \quad \text{which is homogeneous}$$

let $v=tu \Rightarrow \frac{dv}{du} = t+u\frac{dt}{du}$

$$t+u\frac{dt}{du} = -\frac{(u+2tu)}{4tu-2u}$$

$$u\frac{dt}{du} = -\frac{(1+2t)}{(4t-2)} - t$$

$$= \frac{-1-2t-4t^2+2t}{4t-2}$$

$$= -\frac{[4t^2+1]}{2(2t-1)}$$

$$\Rightarrow \frac{2t-1}{4t^2+1}dt = -\frac{1}{2u}du$$

$$\Rightarrow \left[\frac{2t}{4t^2+1} - \frac{1}{4t^2+1}\right]dt = -\frac{1}{2u}du$$

$$\Rightarrow \frac{1}{4}\ln|4t^2+1| + \frac{1}{2}\tan^{-1}(2t) = -\frac{1}{2}\ln u + \frac{1}{2}\ln C$$

$$\Rightarrow \frac{1}{4}\ln\left|4\left(\frac{v}{u}\right)^2+1\right| + \frac{1}{2}\tan^{-1}\left(\frac{2v}{u}\right) = \frac{1}{2}\ln C - \frac{1}{2}\ln u$$

$$\Rightarrow \frac{1}{2}\ln\left|4\left[\frac{y-1}{x-2}\right]^2+1\right| + \tan^{-1}\left(\frac{2y-2}{x-2}\right) = \ln C - \ln|x-2|$$

③ $(x-2y)dx + (2x+y+5)dy = 0$

let $K-2C=0 \Rightarrow K=2C$

$2K+C+5=0$

$\Rightarrow 4C+C+5=0$

$\Rightarrow C=-1$

$\Rightarrow K=-2$

let $x = u-2$, $y = v-$ [redacted]

$dx = du$ $\quad dy = dv$

$(u-2-2v+2)du + (2u-4+v-1+5)dv = 0$

$\Rightarrow \frac{dv}{du} = -\frac{(u-2v)}{2u+v}$ which is homogeneous

let $v = tu \Rightarrow \frac{dv}{dt} = t + u\frac{dt}{du}$

$t + u\frac{dt}{du} = -\frac{(u-2tu)}{2u+tu}$

$= \frac{2t-1}{t+2}$

$\Rightarrow u\frac{dt}{du} = \frac{2t-1-t^2-2t}{t+2}$

$= -\frac{t^2+1}{t+2}$

$\Rightarrow \frac{t+2}{t^2+1}dt = -\frac{1}{u}du$

$\Rightarrow \left(\frac{t}{t^2+1} + \frac{2}{t^2+1}\right)dt = -\frac{1}{u}du$

$\Rightarrow \frac{1}{2}\ln|t^2+1| + 2\tan^{-1}(t) = -\ln u + \ln C$

$$\Rightarrow \frac{1}{2}\ln\left|\left(\frac{v}{u}\right)^2+1\right| + 2\tan^{-1}\left(\frac{v}{u}\right) = -\ln u + \ln C$$

$$\Rightarrow \frac{1}{2}\ln\left|\left(\frac{y+1}{x+2}\right)^2+1\right| + 2\tan^{-1}\left|\frac{y+1}{x+2}\right| = -\ln|x+2| + \ln C$$

8. Riccati Equation

This is an equation of the form

$$\frac{dy}{dx} = A(x)y^2 + B(x)y + C(x) \quad -①$$

where $A(x)$, $B(x)$ and $C(x)$ are continuous functions of x.

The general solution of eqn ① is given by

$$y = y_1 + u \quad -②$$

where y_1 is a particular solution of eqn ①

Substituting ② into ① gives

$$y_1' + u' = A(x)\left[y_1^2 + 2y_1u + u^2\right] + B(x)(y_1+u) + C(x)$$
$$= \left[A(x)y_1^2 + B(x)y_1 + C(x)\right] + A(x)\left[2y_1u + u^2\right] + B(x)u$$

Since y_1 is a particular of eqn ①

$$y_1' = A(x)y_1^2 + B(x)y_1 + C(x)$$

Which means that

$$u' = A(x)\left[2y_1u + u^2\right] + B(x)u$$
$$= A(x)u^2 + \left[2A(x)y_1 + B(x)\right]u$$

which is Bernoulli

Note that the general solution can be found if a particular solution is known. Unfortunately there is no strict algorithm to find a particular solution that depends on $A(x)$, $B(x)$ or $C(x)$,

Examples

solve ① $x^3y' + x^2y - y^2 = 2x^4$, $y_1 = x^2$

② $xy' - 2y = y^2 - x^4$, $y_1 = x^2$

③ $y' + 2xy = 1 + x^2 + y^2$, $y_1 = x$

Soln

Remember to first verify if y_1 is indeed a particular soln

Let $y = x^2 + u$

$\Rightarrow y' = 2x + u'$

$x^3(2x + u') + x^2(x^2 + u) - (x^2 + u)^2 = 2x^4$

$\Rightarrow x^3u' + 2x^4 + x^4 + x^2u - (x^4 + 2x^2u + u^2) = 2x^4$

$\Rightarrow x^3u' + x^2u - 2x^2u - u^2 = 0$

$\Rightarrow x^3u' - x^2u - u^2 = 0$

$\Rightarrow u' - x^{-1}u - u^2x^{-3} = 0$ which is Bernoulli

$\Rightarrow u^{-2}u' - x^{-1}u^{-1} - x^{-3} = 0$

Let $v = -u^{-1}$

$\Rightarrow v' = u^{-2}u'$

$v' + vx^{-1} - x^{-3} = 0$ which is linear

Integrating factor $= \exp\left(\int x^{-1}dx\right)$

$= x$

$xv' + v = x^{-2}$

$\Rightarrow vx = \frac{-1}{x} + C$

$\Rightarrow v = \frac{-1}{x^2} + \frac{C}{x}$

$\Rightarrow \frac{-1}{u} = \frac{-1}{x^2} + \frac{C}{x}$

$$\Rightarrow -\frac{1}{y-x^2} = -\frac{1}{x^2} + \frac{C}{x}$$

$$\Rightarrow y-x^2 = \frac{-x^2}{cx-1}$$

$$\Rightarrow y = \frac{x^2}{1-cx} + x^2 \quad //$$

(2). Let $y = x^2+u$

$$\Rightarrow y' = 2x+u'$$

$$x(2x+u') - 2(x^2+u) = (x^2+u)^2 - x^4$$
$$\Rightarrow 2x^2 + xu' - 2x^2 - 2u = x^4 + 2x^2u + u^2 - x^4$$
$$\Rightarrow xu' - 2u - u^2 - 2x^2u = 0$$
$$\Rightarrow u' - 2ux^{-1} - u^2x^{-1} - 2xu = 0$$
$$\Rightarrow u' - (2x^{-1}+2x)u - u^2x^{-1} = 0 \quad \text{which is Bernoulli}$$
$$\Rightarrow u^{-2}u' - (2x^{-1}+2x)u^{-1} - x^{-1} = 0$$

Let $v = -u^{-1}$

$$\Rightarrow v' = u^{-2}u'$$

$v' + v(2x^{-1}+2x) - x^{-1} = 0$ which is linear

Integrating factor $= \exp\left(\int \left(\frac{2}{x}+2x\right)dx\right)$

$$= e^{\ln x^2 + x^2}$$
$$= x^2 e^{x^2}$$

$$x^2e^{x^2}v' + v(2x^{-1}+2x)x^2e^{x^2} = xe^{x^2}$$
$$\Rightarrow x^2e^{x^2}v = \frac{1}{2}e^{x^2} + C$$

$$\Rightarrow v = \frac{1}{2x^2} + \frac{C}{x^2e^{x^2}}$$

$$\Rightarrow -\frac{1}{u} = \frac{1}{2x^2} + \frac{C}{x^2 e^{x^2}}$$

$$\Rightarrow y = x^2 - \frac{e^{x^2} + C}{2x^2 e^{x^2}}$$

(3) $y' + 2xy = 1 + x^2 + y^2$, $y_1 = x$

let $y = x + u$

$\Rightarrow y' = 1 + u'$

$$1 + u' + 2x(x+u) = 1 + x^2 + (x+u)^2$$

$$\Rightarrow 1 + u' + 2x^2 + 2xu = 1 + x^2 + x^2 + 2xu + u^2$$

$$\Rightarrow u' = u^2$$

$$\Rightarrow u^{-2} du = dx$$

$$\Rightarrow -\frac{1}{u} = x + C$$

$$\Rightarrow \frac{-1}{y - x} = x + C$$

$$\Rightarrow y - x = \frac{-1}{x + C}$$

$$\Rightarrow y = \frac{-1}{x + C} + x$$